

Direct Optimal Truncation of Inverse Harmonic Transseries

Reflection, positive tail densities,
and eventual enveloping

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Abstract

Let $W(X)$ be the positive solution of $\psi(W + \frac{1}{2}) = \log X$, and let $S_N(X) = X + \sum_{n=1}^N h_n X^{1-2n}$ be its direct asymptotic partial sum. The *ProveIt* inverse-harmonic report explicitly leaves open a signed first-omitted-term bound for these sums and their sharp error when $N - \pi X$ is bounded. We resolve the latter question and establish the former on an eventual domain uniform in both X and N .

The key step is an exterior dispersion representation. Digamma reflection identifies the difference between two sectorial inverse branches and produces an eventually positive density $\rho(t) = -2 \operatorname{Re} W_R(it)/\pi \sim 2te^{-2\pi t}$. The inverse is a Stieltjes-type tail integral plus a convergent odd Laurent correction. This representation yields an exact direct remainder and proves that, for suitable X_0, N_0 , every $X \geq X_0$ and $N \geq N_0$ satisfies $0 < (-1)^N (S_N - W) < |h_{N+1}| X^{-2N-1}$.

For $\sigma = N + 1 - \pi X$ bounded, the sharp error is

$$S_N - W = (-1)^N \sqrt{X} e^{-2\pi X} \left\{ 1 + \frac{1}{X} \left[-\frac{\pi}{12} + \frac{2\sigma^2 - 3\sigma + 7/12}{2\pi} \right] + O(X^{-2}) \right\}.$$

We prove an all-orders extension, give the next explicit coefficient, identify the half-next-term law and a square-root truncation window, and quantify the first difference from forward-truncation inversion. Exact rational sign certificates and 190-digit experiments accompany the proofs.

Scope. The thresholds in the uniform enveloping theorem are existential, with an explicit sufficient criterion in analytic constants; no numerical global threshold is asserted. The proofs are not Lean-checked. Priority beyond the inspected repository and the cited literature is not established.

Contents

1	The question, the answer, and its boundary	2
1.1	Two approximations that must not be confused	2
1.2	Main conclusions	2
1.3	What is and is not new in the argument	2
1.4	Notation and quantifiers	3
2	The formal coefficients and the sectorial inverse	3
2.1	Centered coefficients and formal reversion	3
2.2	A local-at-infinity inverse, not global univalence	4
3	Reflection and the positive inverse-jump density	5
3.1	The exact reflected forward map	5
3.2	The density and every algebraic coefficient of its tail	5
4	An exterior dispersion representation	6
4.1	Keeping the bounded part of the continuation problem	6
4.2	Exact moments and the direct remainder	8
5	A uniform eventual enveloping theorem	8
5.1	A reusable positive-tail principle	9
6	Sharp optimal-scale direct truncation	9
6.1	Statement including two explicit correction coefficients	9
6.2	The gamma-expectation reduction	10
6.3	The first correction and its sign	11
6.4	Proof at all orders and the second coefficient	11
7	Large coefficients and wider truncation windows	12
7.1	An independent recovery of the late coefficients	12
7.2	The first-omitted-term fraction on a macroscopic window	13
7.3	A square-root window around optimal truncation	13
8	Comparison with inversion of a forward truncation	14
8.1	The forward integral and its signed remainder	14
9	Finite verification and exact-rational certificates	16
9.1	What was actually computed	16
9.2	Exact sign checks without floating-point digamma values	17
9.3	Reproducibility and limits of the diagnostics	17
10	Consequences for the repository and formalization	18
10.1	The exact gap that has been closed	18
10.2	A tractable dependency order for mechanization	18
11	Further research questions	18
12	Synthesis	20
A	Finite coefficient construction and reproducibility	20
A.1	Computing the density and error coefficients	20
A.2	Files and build instructions	21
B	Provenance and statement-level status	21
B.1	The inspected repository checkpoint	21
B.2	Status ledger	22
	References	23

1 The question, the answer, and its boundary

1.1 Two approximations that must not be confused

For a divergent inverse expansion, truncating and inverting need not commute at exponentially small scales. This distinction is central here. Put

$$F(w) = \psi(w + \tfrac{1}{2}), \quad F(W(X)) = \log X, \quad (1.1)$$

where ψ is the digamma function and X is large and positive. The usual harmonic interpolation $H(x) = \psi(x + 1) + \gamma$ is recovered through

$$H^{-1}(y) = W(e^{y-\gamma}) - \tfrac{1}{2}. \quad (1.2)$$

The centering by one half makes the formal inverse odd:

$$\tilde{W}(X) = X + \sum_{n \geq 1} h_n X^{1-2n}. \quad (1.3)$$

One may either take its direct partial sum S_N , or truncate the forward expansion of F and solve that truncated equation. We denote the latter root by u_N . These are different functions of X even though they have many identical algebraic expansion terms.

The predecessor report, *Exponential Accuracy and Stokes Transport for Inverse Harmonic Transseries* [1], proves a sharp exponential error for u_N , as well as all-order large-coefficient asymptotics and inverse Stokes-sector formulas. Its problem labelled `prob:direct`, entitled “Direct inverse partial sums and an enveloping law”, explicitly asks for the corresponding signed bound and sharp error for S_N . The report correctly warns that positivity of a forward remainder is not a proof of positivity after nonlinear inversion. This article supplies a separate analytic argument for the direct sums.

1.2 Main conclusions

Our exact representation is

$$W(X) = X + E_T(X) - X \int_T^\infty \frac{\rho(t)}{X^2 + t^2} dt, \quad (1.4)$$

where T is sufficiently large, E_T is an odd Laurent series convergent for $|X| > T$, and $\rho(t) > 0$ for $t \geq T$. Importantly, we do not discard E_T . It contains the effect of the bounded part of the continuation geometry and cannot be presumed to vanish.

[Theorems 4.1](#) and [4.3](#) give the representation and the exact remainder. [Theorem 5.1](#) proves a first-omitted-term bound on a rectangle $X \geq X_0$, $N \geq N_0$. [Theorem 6.1](#) gives every algebraic correction to the optimal-scale error. [Theorems 7.2](#) and [7.4](#) describe larger truncation windows. Finally, [corollary 8.2](#) proves

$$S_N(X) - u_N(X) = (-1)^{N+1} \frac{\pi}{6} X^{-1/2} e^{-2\pi X} (1 + O(X^{-1})) \quad (1.5)$$

when $N - \pi X$ is bounded. Thus agreement of the leading exponential constant does not make the two approximation procedures interchangeable.

1.3 What is and is not new in the argument

The coefficient formula for h_n and its fixed-order asymptotic meaning come from the repository’s combinatorial companion [2]. Reflection, the centered digamma expansion, Binet’s formula, and Stirling’s formula are classical [4, 5, 6]. Formal reversion and factorial asymptotics have extensive existing theories;

Fabijonas and Olver [7] and Borinsky [8] are relevant background. Nemes [9] proves enveloping results for reverted cylinder- and Airy-zero expansions by studying analytic inverse functions. That work is a close methodological precedent, not a theorem about the present harmonic inverse.

The contribution developed here is the particular reflected inverse jump, its positive tail density, the retained convergent correction, and the resulting direct-remainder theorems for this family. Our argument does not require a general resurgent-closure theorem. In particular, a positive-real least-term error is not identified with a positive-real Stokes jump: the branch comparison used below is on the imaginary axis. The large-order coefficient result derived from the density is an independent recovery of a result already established in [1], not an additional claim of priority.

The exact proposition answered is a repository research question. Neither its phrasing there nor the targeted literature check establishes that it was a previously named open problem throughout the wider literature. Likewise, the article is a mathematical proof submission with reproducible finite checks, not a declaration of formal verification or independent peer review.

1.4 Notation and quantifiers

We write $A = 2\pi$ throughout. The letter N is the number of retained inverse terms; the first omitted term has index $N + 1$. The bounded-shift parameter is

$$\sigma = N + 1 - \pi X. \quad (1.6)$$

All error bounds involving bounded σ are uniform on any specified compact interval of its values. An all-orders expansion means a separate proved bound for every fixed requested order, not convergence of the resulting formal series.

The real inverse is W ; the right and left sectorial branches are W_R and W_L . Tildes distinguish formal series from functions. E_T is an actual convergent Laurent correction, while $\mathcal{E}_N$ is its tail after N terms. The exact positive direct-remainder integral is denoted by $A_N(X)$; it is not the constant A .

2 The formal coefficients and the sectorial inverse

2.1 Centered coefficients and formal reversion

Define

$$\mathcal{C}(z) = \sum_{j \geq 1} c_j z^j, \quad c_j = -\frac{B_{2j}(1/2)}{2j}. \quad (2.1)$$

The centered digamma expansion, obtained from the standard shifted expansion [6], is

$$F(w) \sim \log w + \mathcal{C}(w^{-2}) \quad (|w| \rightarrow \infty, \quad |\arg w| \leq \theta_0 < \pi). \quad (2.2)$$

The first coefficients are

$$c_1 = \frac{1}{24}, \quad c_2 = -\frac{7}{960}, \quad c_3 = \frac{31}{8064}, \quad c_4 = -\frac{127}{30720}. \quad (2.3)$$

Every fixed truncation has a uniform remainder on a slightly smaller closed sector. Cauchy's formula on disks with radius proportional to $|w|$ then permits termwise differentiation of any such fixed truncation. No assertion about uniformity in an unbounded truncation order is made here.

Proposition 2.1 (The inherited inverse coefficient formula). *There is a unique formal odd Laurent solution of $\log(\tilde{W}/X) + \mathcal{C}(\tilde{W}^{-2}) = 0$. Its coefficients are*

$$h_n = -\frac{1}{2n-1} [z^n] \exp((2n-1)\mathcal{C}(z)), \quad n \geq 1. \quad (2.4)$$

The first five are

$$h_1 = -\frac{1}{24}, \quad h_2 = \frac{3}{640}, \quad h_3 = -\frac{1525}{580608}, \quad h_4 = \frac{615881}{199065600}, \quad h_5 = -\frac{3058641}{504627200}. \quad (2.5)$$

Proof. Put $v = \tilde{W}^{-1}$ and $t = X^{-1}$. The equation becomes $v = t \exp(\mathcal{C}(v^2))$. Formal Lagrange inversion gives its unique odd power-series solution. The Laurent version of the coefficient formula is

$$[t^{2n-1}] \frac{1}{v(t)} = -\frac{1}{2n-1} [u^{2n}] \exp((2n-1)\mathcal{C}(u^2)),$$

which proves (2.4). This identity can also be verified by the formal-residue substitution $t = v \exp(-\mathcal{C}(v^2))$. The displayed rational values follow by finite coefficient extraction. This is the normalization used in the companion [2]. $\square$

2.2 A local-at-infinity inverse, not global univalence

Exponentiating the forward phase removes the logarithm:

$$G(w) = \exp \psi(w + \tfrac{1}{2}). \quad (2.6)$$

It follows from (2.2) that, uniformly on every closed sector strictly inside the cut plane,

$$G(w) = w + O(w^{-1}), \quad G'(w) = 1 + O(w^{-2}), \quad G''(w) = O(w^{-3}). \quad (2.7)$$

The poles of the digamma function on the negative real axis play no role in these sectors.

Lemma 2.2 (The right inverse branch). *Fix $\pi/2 < \theta < \theta_0 < \pi$. For some $R > 0$ there is a holomorphic function W_R on*

$$\mathcal{S}_R = \{z : |z| > R, \quad |\arg z| < \theta\}$$

such that $G(W_R(z)) = z$ and $W_R(z) = z + O(z^{-1})$. It is the unique solution sufficiently close to z , it satisfies

$$W_R(\bar{z}) = \overline{W_R(z)}, \quad W'_R(z) = \frac{1}{G'(W_R(z))} = 1 + O(z^{-2}), \quad (2.8)$$

and, for every fixed m ,

$$W_R(z) = z + \sum_{n=1}^m h_n z^{1-2n} + O_m(|z|^{-2m-1}) \quad (2.9)$$

uniformly on a smaller closed sector. On the positive real axis it is the inverse in (1.1).

Proof. For large z in the specified sector, a disk of radius $K/|z|$ centered at z lies in the larger sector. On its boundary, $G(w) - z = (w - z) + O(|z|^{-1})$. Choose K larger than a uniform constant in this bound. Rouché's theorem then gives exactly one zero in the disk. The derivative estimate in (2.7) makes it a simple zero. The implicit-function theorem gives a holomorphic solution locally in z , and uniqueness in the disks identifies the local solutions on overlaps. This constructs the branch without asserting that G is globally injective.

Conjugation and uniqueness give the first identity in (2.8); differentiation of $G(W_R) = z$ gives the second. Substitution of a fixed formal truncation from [proposition 2.1](#) into G leaves an error $O_m(z^{-2m-1})$. Its actual inverse derivative is uniformly bounded, or equivalently the same small-disk argument applies to the residual. This proves (2.9). Derivatives follow by Cauchy estimates in a slightly smaller sector.

Finally, $F(W_R(z)) - \log z$ is an integer multiple of $2\pi i$, but its asymptotic size is $O(z^{-2})$; hence it is zero. For positive large z the solution is real and positive. The standard trigamma series [5] gives

$$F'(w) = \sum_{k=0}^{\infty} (w + \tfrac{1}{2} + k)^{-2} > 0 \quad (w > 0), \quad (2.10)$$

so this solution agrees with the positive real inverse. $\square$

Define the reflected branch by

$$W_L(z) = -W_R(-z), \quad K(w) = -G(-w). \quad (2.11)$$

Then $K(W_L(z)) = z$. Its sector contains the exterior left half-plane. Both branches are defined near each sufficiently high point of the imaginary axis. Their formal odd expansions agree there; their exact functions need not agree. This is precisely the exponentially small information used next.

3 Reflection and the positive inverse-jump density

3.1 The exact reflected forward map

The digamma reflection identity [4] implies

$$\psi(\tfrac{1}{2} - w) = \psi(\tfrac{1}{2} + w) - \pi \tan(\pi w). \quad (3.1)$$

For $\text{Im } w > 0$ set $q = e^{2\pi i w}$. Since $\tan(\pi w) = i(1 - q)/(1 + q)$, exponentiation gives the exact identity

$$K(w) = G(w) \exp J(w), \quad J(w) = \frac{Aiq}{1 + q}, \quad A = 2\pi. \quad (3.2)$$

The minus sign in the definition of K cancels the factor $\exp(-\pi i) = -1$. Losing that sign would reverse the density below.

Lemma 3.1 (Leading reflected inverse jump). *Let z lie in a fixed-width strip about the positive imaginary axis, and put $t = \text{Im } z \rightarrow +\infty$. Then*

$$W_R(z) - W_L(z) = AizW'_R(z) \exp(AiW_R(z)) + O(t^2 e^{-2At}). \quad (3.3)$$

The error is uniform in each fixed-width strip. The jump and its first derivative are $O(te^{-At})$ there.

Proof. Write $w = W_R(z)$, $v = W_L(z)$, and $\Delta = w - v$. Both w and v are within $O(t^{-1})$ of z , and $q(w) = O(e^{-At})$. By (3.2),

$$K(w) - K(v) = z\{\exp J(w) - 1\}.$$

On the segment between v and w , $K' = 1 + O(t^{-2})$. The integral of this derivative along the segment is bounded away from zero, so the preceding equation first gives $\Delta = O(te^{-At})$. Taylor expansion now gives

$$K'(w)\Delta = z\{\exp J(w) - 1\} + O(t^{-3}\Delta^2).$$

Also $K'(w) = G'(w) + O(te^{-At})$ and $\exp J(w) - 1 = Aiq(w) + O(q(w)^2)$. Division by $G'(w)$, using $1/G'(w) = W'_R(z)$, proves (3.3). All estimates remain valid on a slightly wider strip. Cauchy's derivative estimate on fixed-radius disks then bounds the derivative of the jump by $O(te^{-At})$ as well. $\square$

3.2 The density and every algebraic coefficient of its tail

At $z = it$, conjugation gives $W_L(it) = -\overline{W_R(it)}$. Consequently the jump is real. Define the exact real function

$$\rho(t) = -\frac{W_R(it) - W_L(it)}{\pi} = -\frac{2}{\pi} \text{Re } W_R(it). \quad (3.4)$$

This definition is used only for sufficiently large positive t .

For the algebraic tail coefficients define the formal series

$$\tilde{V}(t) = -i\tilde{W}(it) = t + \sum_{n \geq 1} (-1)^n h_n t^{1-2n}, \quad (3.5)$$

$$\tilde{D}(t) = \tilde{V}'(t) \exp\{-A(\tilde{V}(t) - t)\} = \sum_{j \geq 0} d_j t^{-j}. \quad (3.6)$$

In particular,

$$\tilde{V}(t) = t + \frac{1}{24t} + \frac{3}{640t^3} + \frac{1525}{580608t^5} + \cdots. \quad (3.7)$$

Theorem 3.2 (Positive tail density). *There exists T such that $\rho(t) > 0$ for all $t \geq T$. For every fixed $J \geq 1$,*

$$\rho(t) = 2te^{-At} \left(\sum_{j=0}^{J-1} \frac{d_j}{t^j} + O_J(t^{-J}) \right), \quad (3.8)$$

where the coefficients are exactly those of (3.6). Furthermore $\rho(t), \rho'(t) = O(te^{-At})$. The first coefficients are

$$d_0 = 1, \quad d_1 = -\frac{\pi}{12}, \quad (3.9)$$

$$d_2 = \frac{\pi^2}{288} - \frac{1}{24}, \quad d_3 = -\frac{17\pi}{2880} - \frac{\pi^3}{10368}, \quad (3.10)$$

$$d_4 = -\frac{9}{640} + \frac{11\pi^2}{17280} + \frac{\pi^4}{497664}. \quad (3.11)$$

Proof. At $z = it$, the principal term in (3.3) is $-AtW'_R(it) \exp(AiW_R(it))$. Every fixed algebraic expansion of $-iW_R(it)$ is the corresponding truncation of $\tilde{V}(t)$, and the derivative expansion of $W'_R(it)$ is $\tilde{V}'(t)$. Dividing the jump by $-\pi$ therefore gives (3.8). To justify a requested order, take sufficiently many terms in the uniform expansion (2.9), then use finite Taylor expansions of the exponential and derivative. The error term from (3.3), relative to $2te^{-At}$, is $O(te^{-At})$, smaller than every fixed inverse power of t . No summation of the generally divergent formal series $\tilde{V}$ is used.

The leading coefficient is one, so the exact real density is eventually positive. Smoothness and the derivative bound follow from lemma 3.1. Finally, expand (3.6) using (3.7); this gives (3.9)–(3.11) by finite algebra. $\square$

Remark 3.3 (Why algebraic asymptotics alone would not suffice). Every coefficient of $W_R(it)$ in its inverse-power expansion is purely imaginary. Thus that expansion alone cannot decide the sign of its real part. The exact reflection identity and the comparison of two actual inverse branches are essential. They fix the leading exponentially small real part, and hence the sign of ρ , rather than merely suggesting it from a truncated formal expression.

4 An exterior dispersion representation

4.1 Keeping the bounded part of the continuation problem

Choose T large enough that both sectorial branches exist on the relevant closed half-circles of radius T , and that the density is positive for $t \geq T$. On the exterior right and left half-planes, respectively, set

$$f_R(z) = W_R(z) - z, \quad f_L(z) = W_L(z) - z. \quad (4.1)$$

Write f for this piecewise function, using f_R on the right and f_L on the left. Values at the two cut points of the circle may be assigned arbitrarily for integration. Define

$$E_T(z) = -\frac{1}{2\pi i} \int_{|\zeta|=T}^{\text{ccw}} \frac{f(\zeta)}{\zeta - z} d\zeta, \quad |z| > T. \quad (4.2)$$

This is an ordinary finite contour integral of a bounded piecewise analytic boundary function. It defines a holomorphic function throughout the exterior disk, even though the boundary function itself has two jumps. Let

$$M_T = \sup_{\substack{|\zeta|=T \\ \operatorname{Re} \zeta \neq 0}} |f(\zeta)| < \infty. \quad (4.3)$$

The one-sided boundary values are bounded as well.

Theorem 4.1 (Exterior dispersion with a convergent correction). *For z in either exterior half-plane,*

$$f(z) = E_T(z) - z \int_T^\infty \frac{\rho(t)}{z^2 + t^2} dt. \quad (4.4)$$

The function E_T is odd, has real Laurent coefficients, and has the convergent expansion

$$E_T(z) = \sum_{n \geq 1} e_n z^{1-2n}, \quad |z| > T, \quad (4.5)$$

with

$$|e_n| \leq M_T T^{2n-1}. \quad (4.6)$$

In particular, (1.4) holds for every real $X > T$.

Proof. Apply Cauchy's formula separately to the right and left half-annuli $T < |\zeta| < R$, and add the results. If z is in the right half-annulus, its Cauchy term occurs only in that half-annulus; the left one contributes zero. The same argument works with right and left interchanged.

The inner circular contribution is exactly (4.2). The outer circular contribution tends to zero as $R \rightarrow \infty$: uniformly on its two half-circles, $f(\zeta) = O(R^{-1})$, while $(\zeta - z)^{-1} = O(R^{-1})$ and the total arc length is $2\pi R$. We are left with the two integrals along the imaginary-axis boundaries. Their convergence follows from the exponential jump bound.

For completeness, keep track of their signs. On the upper imaginary axis the right boundary is traversed downward and the left boundary upward. Their sum is

$$-\frac{1}{2\pi i} \int_{iT}^{i\infty} \frac{f_R(\zeta) - f_L(\zeta)}{\zeta - z} d\zeta = \frac{1}{2} \int_T^\infty \frac{\rho(t)}{it - z} dt.$$

Conjugation gives the same real jump $-\pi\rho(t)$ at $-it$. The lower boundary contribution is consequently

$$\frac{1}{2} \int_T^\infty \frac{\rho(t)}{-it - z} dt.$$

Adding the two kernels gives $-z/(z^2 + t^2)$, proving (4.4). This contour argument avoids any assumption that the inverse extends through the bounded disk.

The symmetries $f(-\zeta) = -f(\zeta)$ and $f(\bar{\zeta}) = \overline{f(\zeta)}$ follow from the definitions of the branches. The integral correction has the same symmetries, so E_T is odd and real-symmetric. Alternatively, they follow directly from (4.2). Expanding its kernel for $|z| > T$ gives

$$e_n = \frac{1}{2\pi i} \int_{|\zeta|=T}^{\text{ccw}} f(\zeta) \zeta^{2n-2} d\zeta.$$

The coefficients of the even negative powers vanish by oddness. The arc-length bound gives (4.6) and proves convergence. $\square$

Remark 4.2 (This is not a global Stieltjes assertion). The representation contains a positive Stieltjes-type tail, not a claim that $X - W(X)$ is a Stieltjes transform on its full natural domain. The finite contour correction E_T may be nonzero. Positivity is proved only for the tail density after a sufficiently large cutoff. These two restrictions are exactly what make a local-at-infinity argument enough; a global study of all critical points of the digamma inverse is unnecessary.

The decomposition is not unique in its cutoff. If $T' > T$ is also admissible, then

$$E_{T'}(z) = E_T(z) - z \int_T^{T'} \frac{\rho(t)}{z^2 + t^2} dt \quad (|z| > T'). \quad (4.7)$$

Thus changing the cutoff only moves a compact piece of the integral into a convergent Laurent series. The full function and its asymptotic coefficients are unchanged.

4.2 Exact moments and the direct remainder

All moments

$$\mu_k = \int_T^\infty \rho(t) t^k dt \quad (k \geq 0) \quad (4.8)$$

are finite and positive. Define the convergent correction tail by

$$\mathcal{E}_N(X) = E_T(X) - \sum_{n=1}^N e_n X^{1-2n}. \quad (4.9)$$

For $N = 0$, the empty sum is zero.

Theorem 4.3 (Moment coefficients and an exact direct remainder). *For every $n \geq 1$,*

$$h_n = e_n + (-1)^n \mu_{2n-2}. \quad (4.10)$$

For every $X > T$ and integer $N \geq 0$,

$$S_N(X) - W(X) = (-1)^N A_N(X) - \mathcal{E}_N(X), \quad (4.11)$$

$$A_N(X) = X^{-2N-1} \int_T^\infty \frac{\rho(t) t^{2N}}{1 + (t/X)^2} dt > 0, \quad (4.12)$$

$$|\mathcal{E}_N(X)| \leq \frac{M_T(T/X)^{2N+1}}{1 - (T/X)^2}. \quad (4.13)$$

Proof. Use the finite geometric identity

$$\frac{1}{1+u} = \sum_{k=0}^{N-1} (-u)^k + \frac{(-u)^N}{1+u} \quad (u \geq 0)$$

in the integral of (4.4). Every term is integrable. For fixed N , the remaining integral is bounded by $X^{-2N-1} \mu_{2N}$, so this is also a genuine fixed-order asymptotic expansion. Its coefficients must agree with the unique coefficients in (2.9), giving (4.10). Subtraction gives (4.11) with the stated sign. Summing the geometric majorant (4.6) proves (4.13). $\square$

The exact remainder is the central analytic input that fixed-order reversion did not supply. Unlike a big-O statement with an unspecified constant depending on N , it remains valid with its displayed bound for every N . In particular, it can be analyzed when N grows proportionally to X , or when N and X vary independently.

5 A uniform eventual enveloping theorem

The convergent correction can destroy a global sign argument at small orders. Nevertheless, the moments of a positive unbounded tail eventually outgrow every geometric coefficient bound. The following proof makes that comparison uniform in the evaluation point.

Theorem 5.1 (Eventual enveloping on a two-parameter rectangle). *Let T be an admissible cutoff in theorem 4.1. For every $X_0 > T$ there exists an integer N_0 such that, simultaneously for every $X \geq X_0$ and $N \geq N_0$,*

$$0 < (-1)^N (S_N(X) - W(X)) < |h_{N+1}| X^{-2N-1}. \quad (5.1)$$

In this range, h_{N+1} has sign $(-1)^{N+1}$, and $W(X)$ lies strictly between the consecutive sums $S_N(X)$ and $S_{N+1}(X)$.

Proof. Fix any $L > T$ and let $m = \int_L^{L+1} \rho(t) dt > 0$. For every $X \geq X_0$,

$$A_N(X) \geq X^{-2N-1} \frac{mL^{2N}}{1 + ((L+1)/X_0)^2}. \quad (5.2)$$

The correction bound gives, on the same range,

$$|\mathcal{E}_N(X)| \leq X^{-2N-1} \frac{M_T T^{2N+1}}{1 - (T/X_0)^2}. \quad (5.3)$$

Since $L > T$, the ratio of the right-hand side of (5.3) to that of (5.2) tends to zero geometrically as $N \rightarrow \infty$, independently of X . Choose N_0 so that it is strictly less than one for all $N \geq N_0$. Then (4.11) proves $(-1)^N (S_N - W) > 0$ throughout the rectangle.

Apply the same assertion at $N+1$. The errors of consecutive sums have opposite signs, so W lies strictly between them. Since $S_{N+1} - S_N = h_{N+1} X^{-2N-1}$, their distance is the magnitude of this single term. Each error is strictly smaller than that distance. This proves both the upper bound and the stated coefficient sign. $\square$

Corollary 5.2 (A sufficient threshold criterion). *With X_0, L, m, M_T as in the proof, it suffices to choose N_0 so that*

$$\left(\frac{L}{T}\right)^{2N_0} > \frac{M_T T \{1 + ((L+1)/X_0)^2\}}{m \{1 - (T/X_0)^2\}}. \quad (5.4)$$

The same inequality then holds for all $N \geq N_0$.

The criterion is explicit in analytic constants, but those constants have not been rigorously evaluated here. The theorem is therefore not a numerical statement such as “all $X \geq 1$ and all $N \geq 1$ ”. The six exact-rational examples in section 9 establish selected small cases independently and do not identify X_0 or N_0 .

5.1 A reusable positive-tail principle

The proof of theorem 5.1 used only the exact form (4.11), a geometric bound for the analytic correction, and positive mass on an interval lying beyond its geometric radius. It did not use the specific exponential decay constant 2π . Consequently the same eventual enveloping conclusion holds for any function with an analogous representation and an everywhere positive tail of unbounded support having all moments finite. This principle explains why a bounded collection of continuation complications can be separated from a late-order sign law. It does not show that an arbitrary nonlinear inverse has such a positive representation; establishing that input is the main function-specific task.

6 Sharp optimal-scale direct truncation

6.1 Statement including two explicit correction coefficients

Theorem 6.1 (Sharp direct remainder to all algebraic orders). *Let $X \rightarrow +\infty$ and $N = N(X)$ be an integer with $\sigma = N + 1 - \pi X$ bounded. For each fixed $J \geq 1$ there are polynomials $B_j^-(\sigma)$, $0 \leq j < J$, of degree at most $2j$, such that*

$$S_N(X) - W(X) = (-1)^N \sqrt{X} e^{-2\pi X} \left(\sum_{j=0}^{J-1} \frac{B_j^-(\sigma)}{X^j} + O_J(X^{-J}) \right). \quad (6.1)$$

The error is uniform for σ in any prescribed compact interval. The first coefficients are

$$B_0^-(\sigma) = 1, \quad (6.2)$$

$$B_1^-(\sigma) = -\frac{\pi}{12} + \frac{2\sigma^2 - 3\sigma + 7/12}{2\pi}, \quad (6.3)$$

$$B_2^-(\sigma) = \frac{\pi^2}{288} - \frac{43}{288} - \frac{\sigma^2}{12} + \frac{5\sigma}{24} + \frac{1}{\pi^2} \left(\frac{\sigma^4}{2} - \frac{11\sigma^3}{6} + \frac{5\sigma^2}{3} + \frac{\sigma}{48} - \frac{203}{1152} \right). \quad (6.4)$$

Every further coefficient is determined by a finite algorithm using (3.6), gamma moments, and the shifted Stirling expansion.

6.2 The gamma-expectation reduction

Set

$$b = AX = 2\pi X, \quad s = 2N + 2 = b + 2\sigma, \quad f(v) = \frac{1}{1 + v^2}. \quad (6.5)$$

For $t \geq T$ define the exact positive amplitude

$$D(t) = \frac{\rho(t)}{2te^{-At}}. \quad (6.6)$$

This function has the asymptotic expansion (3.6), but the latter is not used as an infinite convergent series. If U has the gamma density $u^{s-1}e^{-u}/\Gamma(s)$, direct substitution $u = At$ in (4.12) gives the exact identity

$$A_N(X) = 2X \frac{\Gamma(s)}{b^s} \mathbb{E} \left[f(U/b) D(U/A) \mathbf{1}_{\{U \geq AT\}} \right]. \quad (6.7)$$

At the optimal scale, the gamma variable is concentrated near $U = b$, so the relevant part of the density is at $t \approx X$ rather than near the fixed cutoff.

Lemma 6.2 (Uniform finite-order expansion of the expectation). *For bounded $d = s - b$ and fixed J , the expectation in (6.7) has an expansion through b^{-J+1} with remainder $O_J(b^{-J})$. Its coefficients are polynomials in d and finite linear combinations of $A^j d_j$. They can be calculated by Taylor expansion at $v = 1$ and the exact identities*

$$\mathbb{E}(U/b - 1)^k = b^{-k} \sum_{\ell=0}^k \binom{k}{\ell} (-b)^{k-\ell} (s)_\ell. \quad (6.8)$$

The coefficient of b^{-r} has degree at most $2r$ in d .

Proof. The amplitude D is bounded on $[T, \infty)$ and, for each fixed J ,

$$D(t) = \sum_{j=0}^{J-1} d_j t^{-j} + O_J(t^{-J}).$$

On $U \geq b/2$, inserting this expansion leaves a remainder $O_J(b^{-J})$ in the expectation. The part $AT \leq U < b/2$ is exponentially small in b , uniformly for bounded d , by a gamma Chernoff bound. The same is true after multiplication by any fixed inverse power $(U/b)^{-j}$: for example, use Cauchy–Schwarz together with

$$\mathbb{E}(U/b)^{-2j} = b^{2j} \frac{\Gamma(s-2j)}{\Gamma(s)} = O_j(1)$$

and a gamma lower-tail bound, once $s > 2j$. The omitted interval $0 < U < AT$ is smaller still and affects no algebraic coefficient. These estimates justify replacing the cutoff expectation, to any fixed algebraic order, by

$$\sum_{j=0}^{J-1} d_j A^j b^{-j} \mathbb{E}\{f(U/b)(U/b)^{-j}\}.$$

For each summand, Taylor-expand $g_j(v) = f(v)v^{-j}$ at one on $1/2 \leq v \leq 3/2$. On this compact interval all required derivatives are bounded. Gamma upper and lower tails outside it are exponentially small; the preceding negative-moment estimate also controls the lower tail for $j > 0$. Taylor's theorem through degree $2J - 1$, and the even moment $\mathbb{E}(U/b - 1)^{2J} = O_J(b^{-J})$, give a remainder of the required order. Taking a few more terms when multiplied by a known power of b^{-1} is harmless. Identity (6.8) follows by the binomial theorem and $\mathbb{E}U^\ell = (s)_\ell$.

For the degree bound, write $U/b - 1 = (U - s)/b + d/b$. A centered gamma moment of order k is a polynomial in s of degree at most $\lfloor k/2 \rfloor$ for $k \geq 2$, with the first centered moment zero. This follows directly by expanding the centered moment generating function $\exp\{s \sum_{r \geq 2} t^r/r\}$. Consequently a coefficient at b^{-r} in the shifted normalized moments can involve at most $2r$ powers of d . The finite Taylor operations and the additional factors b^{-j} preserve the asserted bound. $\square$

6.3 The first correction and its sign

For $f(v) = (1 + v^2)^{-1}$,

$$f(1) = \frac{1}{2}, \quad f'(1) = -\frac{1}{2}, \quad f''(1) = \frac{1}{2}, \quad f'''(1) = 0.$$

With $d = 2\sigma$, the gamma moments give

$$\mathbb{E}f(U/b) = \frac{1}{2} + \frac{1/4 - \sigma}{b} + O(b^{-2}). \quad (6.9)$$

The density amplitude contributes $d_1 A b^{-1} \mathbb{E}\{f(U/b)(U/b)^{-1}\} = d_1 A/(2b) + O(b^{-2})$. Thus the expectation in (6.7) equals

$$\frac{1}{2} + \frac{1/4 - \sigma + A d_1/2}{b} + O(b^{-2}). \quad (6.10)$$

The shifted Stirling expansion [6], uniformly for bounded d , gives

$$\frac{\Gamma(b+d)}{b^{b+d}} = \sqrt{2\pi} b^{-1/2} e^{-b} \left(1 + \frac{d^2/2 - d/2 + 1/12}{b} + O(b^{-2}) \right). \quad (6.11)$$

Since $2X\sqrt{2\pi} b^{-1/2} = 2\sqrt{X}$, multiplying (6.10) and (6.11) yields

$$A_N(X) = \sqrt{X} e^{-AX} \left(1 + \frac{2\sigma^2 - 3\sigma + 7/12}{AX} + \frac{d_1}{X} + O(X^{-2}) \right).$$

The value $d_1 = -\pi/12$ explains the minus sign in (6.3). There is no substitution of a forward-truncation error in this calculation.

6.4 Proof at all orders and the second coefficient

Proof of theorem 6.1. For bounded σ , the convergent correction is negligible to every algebraic order relative to the first exponential scale. Indeed, (4.13) and $2N + 1 = AX + O(1)$ imply

$$|\mathcal{E}_N(X)| \leq \exp\{-AX \log X + O(X)\} = o(\sqrt{X} e^{-AX} X^{-J}) \quad (6.12)$$

for every fixed J . It therefore suffices to expand (6.7). Use lemma 6.2 and the finite shifted logarithmic Stirling expansion

$$\log \frac{\Gamma(b+d)}{\sqrt{2\pi}e^{-b}b^{b+d-1/2}} \sim \sum_{r \geq 1} \frac{(-1)^{r+1}B_{r+1}(d)}{r(r+1)b^r}. \quad (6.13)$$

Every fixed truncation is uniform for d in a compact interval [6]. Expanding its exponential and multiplying the two finite expansions proves (6.1). Its coefficient at b^{-r} has degree at most $2r$ in d : a Stirling-logarithm coefficient has degree $r+1 \leq 2r$, and the expectation has the same bound by the lemma. Products preserve the bound. Replacing b by AX and d by 2σ proves the stated polynomial degree bound and the finite algorithm.

Here are enough intermediate quantities to audit the second coefficient. Taylor expansion and the exact moments give

$$\begin{aligned} \mathbb{E}f(U/b) &= \frac{1}{2} + \frac{1/4 - \sigma}{b} + \frac{\sigma^2 + \sigma/2 - 3/8}{b^2} + O(b^{-3}), \\ \mathbb{E}\{f(U/b)(U/b)^{-1}\} &= \frac{1}{2} + \frac{5/4 - 2\sigma}{b} + O(b^{-2}), \\ \mathbb{E}\{f(U/b)(U/b)^{-2}\} &= \frac{1}{2} + O(b^{-1}). \end{aligned}$$

Put $L_1 = B_2(2\sigma)/2$ and $L_2 = -B_3(2\sigma)/6$, where the B_k on this line are Bernoulli polynomials, not the error coefficients B_k^- . The Stirling multiplier through b^{-2} is $1 + L_1/b + (L_2 + L_1^2/2)/b^2$. The expectation multiplier through b^{-2} has coefficients

$$\begin{aligned} Q_1 &= \frac{1}{4} - \sigma + Ad_1/2, \\ Q_2 &= \sigma^2 + \sigma/2 - \frac{3}{8} + Ad_1(\frac{5}{4} - 2\sigma) + A^2d_2/2. \end{aligned}$$

Therefore

$$B_2^-(\sigma) = \frac{2Q_2 + 2L_1Q_1 + L_2 + L_1^2/2}{A^2}.$$

Inserting (3.9)–(3.10) and simplifying proves (6.4). These finite simplifications are also checked by the included symbolic program. $\square$

Remark 6.3 (Optimal scale versus an exact minimizing integer). The theorem covers every integer choice with $N - \pi X$ bounded. The continuous minimum of $B_1^-(\sigma)$ occurs at $\sigma = 3/4$, suggesting N close to $\pi X - 1/4$. This is an asymptotic guide. The article does not claim that this rounded prescription is the exact best integer for every X , especially near a tie between neighboring integers.

7 Large coefficients and wider truncation windows

7.1 An independent recovery of the late coefficients

The same density determines both the coefficients and the direct error. This provides a useful consistency test against the independent factorial-convolution proof in the predecessor report.

Corollary 7.1 (Moment form of the complete late-coefficient expansion). *For each fixed $J \geq 1$,*

$$h_n = \frac{2(-1)^n\Gamma(2n)}{A^{2n}} \left\{ \sum_{j=0}^{J-1} \frac{d_j A^j}{(2n-1)(2n-2)\cdots(2n-j)} + O_J(n^{-J}) \right\}, \quad (7.1)$$

where the denominator of the $j = 0$ term is one. In particular,

$$h_n \sim \frac{2(-1)^n\Gamma(2n)}{(2\pi)^{2n}}. \quad (7.2)$$

Proof. Insert (3.8) into the moment in (4.10). For n sufficiently large, each term can be integrated over $(0, \infty)$ instead of (T, ∞) with an error at most geometric in n . The resulting integrals are $\Gamma(2n - j)/A^{2n-j}$. The amplitude remainder is bounded by a constant times $\Gamma(2n - J)/A^{2n-J}$. The coefficient e_n and the missing compact integrals are negligible relative to this factorial scale. Dividing by $\Gamma(2n)/A^{2n}$ proves the assertion. $\square$

Expanding the first correction gives $h_n/[2(-1)^n\Gamma(2n)/A^{2n}] = 1 - \pi^2/(12n) + O(n^{-2})$. This agrees with the large-order normalization in [1]. The point here is the analytic connection to (4.11), not the rediscovery of the leading factorial coefficient.

7.2 The first-omitted-term fraction on a macroscopic window

Theorem 7.2 (A uniform truncation-ratio profile). *Let*

$$r = \frac{N+1}{\pi X}. \quad (7.3)$$

As $X \rightarrow \infty$, uniformly when r stays in any compact subinterval of $(0, \infty)$,

$$\frac{(-1)^N \{S_N(X) - W(X)\}}{|h_{N+1}|X^{-2N-1}} = \frac{1}{1+r^2} + O(X^{-1}). \quad (7.4)$$

Proof. In this range N is comparable to X . Both the coefficient correction e_{N+1} and the remainder $\mathcal{E}_N$ are negligible compared with the positive moment scales, uniformly in r , by (4.6) and Stirling's formula. Thus the ratio differs negligibly from

$$\frac{\int_T^\infty \rho(t)t^{2N}f(t/X)dt}{\int_T^\infty \rho(t)t^{2N}dt}. \quad (7.5)$$

Under $u = At$, the leading weight is gamma with shape $s = 2N + 2 = br$. Its normalized mean is $\mathbb{E}(U/b) = r$, and $\text{Var}(U/b) = r/b = O(X^{-1})$. The amplitude is $D(U/A) = 1 + O(X^{-1})$ on a concentration interval about r , uniformly when r stays in a fixed compact subset of $(0, \infty)$. Outside such an interval the gamma tails are exponentially small; fixed negative moments control the small- U part as in lemma 6.2. Taylor's theorem for f , whose second derivative is bounded on the positive axis, gives $\mathbb{E}f(U/b) = f(r) + O(X^{-1})$. The amplitude affects numerator and denominator by $O(X^{-1})$, so (7.5) equals $f(r) + O(X^{-1})$. $\square$

Corollary 7.3 (Half the first omitted term at optimal scale). *If $N - \pi X$ is bounded, then*

$$\frac{(-1)^N(S_N - W)}{|h_{N+1}|X^{-2N-1}} = \frac{1}{2} + O(X^{-1}). \quad (7.6)$$

In the same regime,

$$|h_{N+1}|X^{-2N-1} = 2\sqrt{X}e^{-2\pi X}(1 + O(X^{-1})). \quad (7.7)$$

The coefficient two in the least-term size and the coefficient one in the actual error thus have a precise explanation: the remainder kernel is $(1 + (t/X)^2)^{-1}$, and its mass is concentrated at $t/X \approx 1$. It is not necessary to infer this factor from a heuristic rule about alternating divergent series.

7.3 A square-root window around optimal truncation

Theorem 7.4 (The square-root truncation profile). *Put*

$$\tau = \frac{N+1-\pi X}{\sqrt{X}}. \quad (7.8)$$

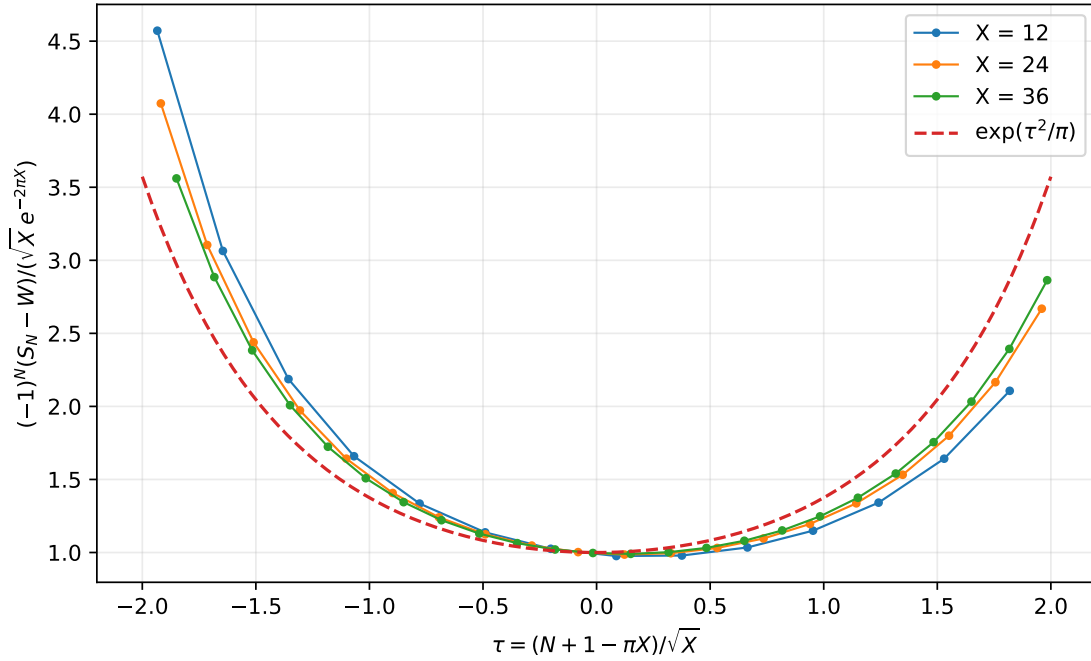


Figure 1: Direct inverse errors across a square-root truncation window. The solid curves join values calculated at integer N with 190-digit arithmetic. The dashed curve is the limiting profile in [theorem 7.4](#). Visible left–right asymmetry at finite X is compatible with the stated $O(X^{-1/2})$ relative correction. This is a numerical illustration, not a substitute for the uniform proof.

For every fixed $K > 0$, uniformly for $|\tau| \leq K$,

$$\frac{(-1)^N \{S_N(X) - W(X)\}}{\sqrt{X} e^{-2\pi X}} = \exp(\tau^2/\pi) (1 + O_K(X^{-1/2})). \quad (7.9)$$

Proof. Now $s = b + d$ with $d = 2\tau\sqrt{X} = O_K(\sqrt{b})$. Stirling’s formula expanded with this growing shift gives

$$\frac{\Gamma(b + d)}{b^{b+d}} = \sqrt{2\pi} b^{-1/2} e^{-b} \exp\left(\frac{d^2}{2b}\right) (1 + O_K(b^{-1/2})). \quad (7.10)$$

Indeed, expand $(b + d - 1/2) \log(1 + d/b) - d$; the leading term is $d^2/(2b)$, while the terms $-d/(2b)$ and d^3/b^2 are $O_K(b^{-1/2})$. The usual Stirling remainder is smaller.

The gamma variable U/b has mean $1 + O_K(b^{-1/2})$ and variance $O_K(b^{-1})$. Hence the expectation in (6.7) is $1/2 + O_K(b^{-1/2})$; the factor D changes it only by $O_K(b^{-1})$. The fixed cutoff and convergent correction are negligible as before. Finally, $d^2/(2b) = \tau^2/\pi$, proving (7.9). $\square$

8 Comparison with inversion of a forward truncation

8.1 The forward integral and its signed remainder

To make the comparison self-contained, we recall the exact forward integral and derive the correction needed here. Binet’s digamma formula and duplication [4, 5] give, for $w > 0$,

$$C(w) := F(w) - \log w = 2 \int_0^\infty \frac{t}{w^2 + t^2} \frac{dt}{e^{2\pi t} + 1}. \quad (8.1)$$

One direct derivation starts with the Bose-kernel form

$$\psi(w) = \log w - \frac{1}{2w} - 2 \int_0^\infty \frac{t}{w^2 + t^2} \frac{dt}{e^{2\pi t} - 1}$$

and substitutes it into $\psi(w + 1/2) = 2\psi(2w) - \psi(w) - 2 \log 2$. After rescaling the integral with $2w$, the kernel identity $(e^u - 1)^{-1} - 2(e^{2u} - 1)^{-1} = (e^u + 1)^{-1}$ yields (8.1).

Write

$$P_N(w) = \log w + \sum_{j=1}^N c_j w^{-2j}, \quad R_N(w) = F(w) - P_N(w). \quad (8.2)$$

Finite geometric expansion of (8.1) gives

$$R_N(w) = 2(-1)^N w^{-2N-2} \int_0^\infty \frac{t^{2N+1}}{1 + (t/w)^2} \frac{dt}{e^{At} + 1}, \quad (8.3)$$

$$0 < (-1)^N R_N(w) < |c_{N+1}| w^{-2N-2}. \quad (8.4)$$

The moment formula identifies the coefficients with (2.1). The derivative of this exact remainder is bounded in magnitude by $(2N + 2)|R_N(w)|/w$: use the equivalent form with prefactor w^{-2N} and denominator $w^2 + t^2$, then differentiate under the positive integral. A larger bound with the first omitted term in place of $|R_N|$ is therefore valid as well.

Proposition 8.1 (Forward-truncation inverse at the same scale). *For $N = \pi X$ bounded and X sufficiently large, the equation $P_N(u_N) = \log X$ has a unique solution near $W(X)$, and*

$$u_N - W = (-1)^N \sqrt{X} e^{-AX} \left(1 + \frac{B_1^+(\sigma)}{X} + O(X^{-2}) \right), \quad (8.5)$$

where

$$B_1^+(\sigma) = \frac{\pi}{12} + \frac{2\sigma^2 - 3\sigma + 7/12}{2\pi}. \quad (8.6)$$

This is the first-correction theorem of the predecessor report [1] in the present notation.

Proof. On an interval $w = X + O(X^{-1})$ containing $W(X)$, the first omitted forward term has size $O(X^{-1/2}e^{-AX})$. This follows from its explicit Bernoulli moment formula and Stirling's formula. The derivative bound above shows that $P'_N = F' - R'_N$ is positive and asymptotic to X^{-1} there. Its residual at W is $-R_N(W)$. The intermediate value theorem therefore produces a unique local root at distance $O(\sqrt{X}e^{-AX})$. Taylor's theorem gives

$$u_N - W = \frac{R_N(W)}{F'(W)} + O(Xe^{-2AX}). \quad (8.7)$$

For example, $F'' = O(X^{-2})$ and $R'_N = O(X^{-1/2}e^{-AX})$ give the displayed error after division by $F'(W) \asymp X^{-1}$.

In (8.3), replacing $(e^{At} + 1)^{-1}$ by e^{-At} changes the normalized moment by at most 2^{-2N-2} . The same gamma calculation as (6.9)–(6.11), now with $b_w = Aw$ and $\eta = N + 1 - \pi w$, gives

$$R_N(w) = (-1)^N w^{-1/2} e^{-Aw} \left(1 + \frac{2\eta^2 - 3\eta + 7/12}{Aw} + O(w^{-2}) \right).$$

There is no density coefficient d_1 in this forward formula. Finally,

$$W = X - \frac{1}{24X} + O(X^{-3}), \quad \frac{1}{F'(W)} = W\{1 + O(W^{-2})\}.$$

The substitution into (8.7) introduces $e^{-A(W-X)} = 1 + \pi/(12X) + O(X^{-2})$. This yields (8.5)–(8.6). $\square$

X	N	Normalized error	Through B_2	Direct/forward ratio
4	12	0.9180683138	0.9185311808	1.04815277
8	25	0.9573226104	0.9573811152	1.02006614
12	37	0.9758754944	0.9758879980	1.01823786
20	62	0.9877449175	0.9877479997	1.01309911
30	94	0.9884110336	0.9884119587	1.00511017
40	125	0.9926247019	0.9926249934	1.00518857

Table 1: Numerical optimal-scale checks at $N = \lfloor \pi X \rfloor$. The normalized error is $(-1)^N (S_N - W)/(\sqrt{X}e^{-2\pi X})$. The next column is $1 + B_1^-(\sigma)/X + B_2^-(\sigma)/X^2$. The last column is $(-1)^{N+1} (S_N - u_N)/(\pi X^{-1/2}e^{-2\pi X}/6)$ and tends to one. Displayed decimals are rounded diagnostics, not interval endpoints.

Corollary 8.2 (The first discrepancy between the two procedures). *Uniformly for bounded $\sigma = N + 1 - \pi X$,*

$$S_N - u_N = (-1)^{N+1} \frac{\pi}{6} X^{-1/2} e^{-2\pi X} (1 + O(X^{-1})). \quad (8.8)$$

Proof. Subtract (8.5) from (6.1) through the first correction. The leading constants cancel, while $B_1^- - B_1^+ = -\pi/6$. The uniform errors are one inverse power smaller than the nonzero difference. $\square$

The direct correction and the forward correction arise from different analytic mechanisms. The direct one contains the tail amplitude of the reflected inverse jump. The forward one contains the exponential effect of evaluating the forward remainder at $W(X) = X - 1/(24X) + \dots$. The opposite signs are not an arbitrary normalization choice.

9 Finite verification and exact-rational certificates

9.1 What was actually computed

The accompanying program `verification/verify.py` performs finite exact arithmetic checks and separate high-precision analytic diagnostics. It generates 200 inverse coefficients numerically from the finite exponential recurrence, computes the first 20 independently as exact rationals, and checks the formal inverse residual through degree eight. It then evaluates direct errors, density tails, macroscopic ratios, and the square-root window at a total of 76 recorded parameter choices. The main run uses 190 decimal digits. All programmed assertions passed.

For a fixed n , the coefficient calculation uses $E_0 = 1$ and

$$E_k = \frac{2n-1}{k} \sum_{j=1}^k j c_j E_{k-j}, \quad h_n = -\frac{E_n}{2n-1}. \quad (9.1)$$

This is coefficient comparison in the derivative of $\exp((2n-1)\mathcal{C}(z))$ and avoids symbolic expansion of a large exponential. The exact residual check independently substitutes $1 + \sum_{n=1}^8 h_n z^n$ into the normalized logarithmic inverse equation.

At $X = 40$, the normalized direct error is approximately 0.99262470191319, while the expansion through B_2^- gives 0.99262499343886. The difference multiplied by X^3 is approximately -0.01865764 , consistent with the next bounded coefficient. Because σ changes with X , this last quantity is not expected to converge to one constant along the displayed table.

The imaginary-axis density is particularly sensitive to precision: $W_R(it)$ has an imaginary part of size t but a real part of size $te^{-2\pi t}$. At $t = 40$, the computed density is approximately $5.6250672202997191 \cdot 10^{-108}$. The program uses the defining complex digamma equation at high precision and independently

checks (3.2). These checks support the sign and normalization; the proof of positivity for all sufficiently large t is theorem 3.2, not the finite data.

9.2 Exact sign checks without floating-point digamma values

A rational candidate $v > 0$ can be tested using only rational arithmetic and the proven analytic bound (8.4). Shift by a positive integer L :

$$F(v) - \log X = \log \frac{v+L}{X} + \sum_{j=1}^K c_j (v+L)^{-2j} - \sum_{k=0}^{L-1} \frac{1}{v+1/2+k} + R_K(v+L). \quad (9.2)$$

The recurrence for the digamma function justifies the shift. It makes the forward remainder much smaller than the direct error being tested, without requiring the latter's asymptotic formula as an input.

For any rational $r > 0$, put $t = (r-1)/(r+1)$. Then

$$\log r = 2 \sum_{k=0}^{m-1} \frac{t^{2k+1}}{2k+1} + \varepsilon_m, \quad (9.3)$$

$$|\varepsilon_m| \leq \frac{2|t|^{2m+1}}{(2m+1)(1-t^2)}. \quad (9.4)$$

The bound follows from the tail of the absolutely convergent series for $2 \operatorname{arctanh} t$. Every endpoint in the resulting enclosure for (9.2) is rational when X, v are rational. For even K , the final forward remainder lies strictly between zero and $|c_{K+1}|(v+L)^{-2K-2}$. Thus the enclosure has a rigorously checkable sign.

The recorded certificates use $L = 4$, $K = 24$, and $m = 100$, at the exact rational candidates $v = S_N(X)$. They prove the six signs in the following three pairs:

X	Exactly certified ordering
2	$S_7(2) < W(2) < S_6(2)$
3	$S_9(3) < W(3) < S_{10}(3)$
4	$S_{13}(4) < W(4) < S_{12}(4)$

The stored `certificates.json` contains the exact rational candidates and exact lower and upper bounds for their forward residuals. The program checks each sign as an integer comparison, not by converting the fractions to floating point. Strict monotonicity (2.10) transfers each sign to the inverse ordering. The corresponding first-omitted-term bounds follow from the distance between consecutive sums.

These finite certificates are independent of the unknown eventual thresholds. They are also not proof-assistant kernel certificates: their logical trust boundary is the mathematical integral inequality plus the implementation of exact integer and rational arithmetic.

9.3 Reproducibility and limits of the diagnostics

The JSON output separates exact identities, exact sign enclosures, and floating-point values. The numerical checks use the actual inverse of the digamma function, not an optimally truncated series as a reference. The large cancellation in $S_N - W$ and in $\operatorname{Re} W_R(it)$ is the reason for the high working precision. The program permits a higher precision run and rejects a precision below 160 digits for this data set.

The figure and the table can be rebuilt from the program. They do not establish uniform convergence, determine exact threshold constants, or certify numerical values over a continuum. Those statements belong to the human proofs or, for the thresholds, to the further work below.

10 Consequences for the repository and formalization

10.1 The exact gap that has been closed

The predecessor’s `prob:direct` has two parts: a useful signed first-omitted-term domain, and a sharp direct error at bounded $N - \pi X$. Theorems 5.1 and 6.1 answer those parts, respectively. The first domain is an eventual rectangle with existential constants; the second result includes all algebraic corrections to the first exponential scale. No substitution of u_N for S_N is made.

The new analytic representation can be placed after the predecessor’s section on inverse Stokes transport and before its further questions. The coefficient formula and the late-coefficient theorem need not be copied as new contributions when merging. The present proof of late coefficients is best retained as an analytic cross-check or alternative proof. The two error corrections B_1^+ and B_1^- should retain their superscripts: collapsing the notation would erase the main distinction.

10.2 A tractable dependency order for mechanization

The repository’s transseries development includes asymptotic scales, flatness, well-based supports, and differential blocks; its README is careful about the limits of its formal crosswalk [3]. Nothing here changes those status declarations automatically.

A sensible first formal target is the abstract tail comparison: assuming (4.11), a positive mass interval, and a geometric correction bound, prove theorem 5.1. This is a real inequality argument and does not require a digamma continuation library. The finite geometric identity, exact coefficient recurrence, and rational logarithm bound are other small, independent targets.

The next layer is theorem 4.3, conditional on the dispersion representation. It requires integrability of the positive moments but no asymptotic probability theory. The complex-analytic layer then comprises the sectorial inverse construction, reflection, and the two-half-annulus Cauchy argument. Only the final sharp-amplitude layer needs the uniform gamma concentration estimates and shifted Stirling remainders.

This order exposes assumptions instead of hiding them in a nominal “transseries” structure. In particular, a formal support theorem does not imply the existence of the analytic density, and a checked finite coefficient identity does not imply a uniform optimal-truncation theorem. No new Lean source is included or claimed to have been compiled.

11 Further research questions

The following questions are not premises left unfinished in the proofs above. They concern stronger domains, constructive bounds, or genuinely additional analytic information. Their inclusion does not assert that every formulation is unknown in the wider literature.

Research question 1 (Effective thresholds and the full positive-axis envelope). Determine explicit numerical T , X_0 , N_0 for theorem 5.1, and then decide how far the strict bound extends toward small X and small N .

One route is to bound $G(w) - w$ and its derivatives explicitly on chosen sectorial disks, certify the sign of the reflected jump, and estimate M_T and one tail mass m . Criterion (5.4) would then produce a rigorous, possibly nonoptimal threshold. A separate finite certification step could cover the omitted compact range. A statement for every positive point of the inverse’s domain would require more than the current local-at-infinity proof.

Research question 2 (Minimal cutoff and bounded inverse geometry). Identify the analytic obstructions that determine the smallest useful circle in (4.2), and describe the relevant inverse branch points arising from $\psi'(w + 1/2) = 0$.

The present proof deliberately avoids locating them. A sharper geometric analysis could lower the convergence radius of E_T , improve the constants in the envelope criterion, and distinguish genuine inverse singularities from artifacts of the chosen gluing contour. It must keep track of the chosen inverse sheet, not just list zeros of the derivative.

Research question 3 (A constructive dispersion evaluator). Develop a rigorously bounded algorithm that evaluates E_T and the positive tail integral separately and efficiently.

The exact representation is not yet a fast quadrature algorithm. The imaginary-axis density is exponentially small and naive evaluation loses many digits. Reflection-based formulas that compute the jump directly, without subtracting nearly opposite inverse branches, should improve conditioning. Certified contour quadrature for E_T and positive quadrature for the tail would then give a new numerical inverse method with an explicit error budget.

Research question 4 (Beyond the full first exponential block). Construct a direct hyperasymptotic approximation whose remainder is quantified beyond the complete $e^{-2\pi X}$ block.

Subtracting finitely many terms B_j^-/X^j does not reach a second exponential scale. One must resolve the remainder of the density amplitude itself and the nonlinear higher reflected-jump sectors. The $O(t^2 e^{-4\pi t})$ bound in lemma 3.1 is a starting estimate, not by itself a full second-level hyperasymptotic theorem.

Research question 5 (Local Borel singularities from the dispersion density). Recover explicit singular germs of the inverse Borel transform at the nearest imaginary actions and compare them with the all-sector transport formulas in the predecessor report.

The density fixes complete factorial coefficient expansions, but local Borel singularity statements need specified continuation sheets and control of analytic remainders. Matching those statements to the exact reflection jump would provide a constructive link between the two analytic descriptions rather than just agreement of their leading constants.

Research question 6 (Complex-direction optimal truncation). Determine uniform direct remainder formulas as X rotates toward the imaginary directions used in the branch comparison.

For positive X , the denominator $X^2 + t^2$ has no positive-real pole. After rotation, its poles can approach the integration path. A uniform transition analysis must specify the lateral contour and track residues. It should distinguish changes in the optimally truncated remainder from actual differences of sectorial sums.

Research question 7 (Uniform generalized-harmonic inversion near order one). Extend the dispersion method to a joint limit involving generalized harmonic functions of order $s \downarrow 1$ and $X \rightarrow \infty$.

The natural inverse coordinates for fixed $s > 1$ are singular at $s = 1$. A successful extension needs a common target normalization and a parameter-uniform analogue of the reflected density. A fixed- s expansion followed by formal substitution $s = 1$ cannot supply these bounds.

Research question 8 (Which inverse families have an eventually positive jump?). Find checkable sufficient conditions on a centered phase and its reflection or functional relation that imply a representation of the form (4.4) with an eventually one-signed density.

The elementary envelope principle in section 5 is broadly applicable once its hypotheses are known. The nontrivial classification problem is to establish those hypotheses. Competing conjugate actions can create oscillatory densities or cancellations, so nonlinear inversion should not be expected to preserve positivity without a structural reason.

Research question 9 (Exact best truncation and complexity). Determine the exact minimizing integer N away from, and near, the transition between two adjacent minimizers, and obtain a bit-complexity bound for a certified evaluation to a requested tolerance.

The polynomial coefficients in [theorem 6.1](#) give a systematic way to refine the continuous center $\sigma = 3/4$. A uniform treatment near discrete ties needs a bound smaller than the difference of adjacent errors. On the computational side, fast series arithmetic and binary splitting may improve the transparent recurrence used here, but a complexity claim must account for rational numerator growth and the precision needed to resolve exponentially small quantities.

12 Synthesis

The missing direct-truncation estimate is supplied by an analytic representation of the inverse, not by further formal reversion. Reflection fixes a small difference of actual inverse branches; that difference gives a positive density on a sufficiently distant tail; Cauchy's formula separates the tail from a convergent finite-contour correction. The resulting exact remainder permits both an independent large-order sign argument and a uniform saddle-concentration calculation.

The conclusions have different strengths and different boundaries. The enveloping theorem is uniform in two independent large parameters, but its threshold constants are not numerically certified here. The optimal-scale theorem has an explicit leading constant, two displayed correction coefficients, and a proved all-orders construction. The macroscopic and square-root profiles explain how the error varies as the truncation index moves. Finally, the direct and forward procedures agree at leading exponential order and disagree at the next one by the nonzero constant $\pi/6$.

These results answer the precisely identified repository question while leaving the global inverse geometry, effective thresholds, higher exponential layers, and wider families as distinct research tasks.

A Finite coefficient construction and reproducibility

A.1 Computing the density and error coefficients

To determine $d_0, \dots, d_K$, use only the coefficients h_n with $2n - 1 \leq K$ in (3.5), differentiate the finite series, and expand (3.6) through t^{-K} . All these operations take place in a finite polynomial ring with rational coefficients and powers of π . For example, if $a_1 = 1/24$ and $a_2 = 3/640$, then

$$\tilde{V}'(t) = 1 - a_1 t^{-2} - 3a_2 t^{-4} + \dots, \quad \exp\{-A(\tilde{V} - t)\} = \exp(-Aa_1 t^{-1} - Aa_2 t^{-3} - \dots).$$

This immediately produces (3.9)–(3.11).

For a requested error coefficient B_K^- , introduce a formal variable $e = b^{-1}$ and $d = 2\sigma$. Expand the finite expression

$$2 \exp \left\{ \sum_{r=1}^K \frac{(-1)^{r+1} B_{r+1}(d)}{r(r+1)} e^r \right\} \sum_{j=0}^K d_j A^j e^j \sum_{k=0}^{2K} \frac{g_j^{(k)}(1)}{k!} \mathcal{M}_k(e, d), \quad (\text{A.1})$$

where $g_j(v) = (1 + v^2)^{-1} v^{-j}$ and

$$\mathcal{M}_k(e, d) = e^k \sum_{\ell=0}^k \binom{k}{\ell} (-e^{-1})^{k-\ell} (e^{-1} + d)^\ell. \quad (\text{A.2})$$

After cancellation, $\mathcal{M}_k$ is a polynomial in e . The coefficient of e^r in (A.1), divided by A^r , is $B_r^-(\sigma)$ for $r \leq K$. Terms that cannot contribute through e^K may be discarded at every multiplication. The proof in [lemma 6.2](#) justifies the algorithm's analytic meaning; the algorithm alone would not prove its remainder bound.

A.2 Files and build instructions

The package contains the editable source, the compiled PDF, the stored figure, verification programs, exact rational certificates, and the recorded numerical and symbolic outputs. From the package directory run

```
python verification/verify.py
python verification/derive_amplitudes.py
pdflatex -interaction=nonstopmode -halt-on-error \
  direct_inverse_harmonic.tex
pdflatex -interaction=nonstopmode -halt-on-error \
  direct_inverse_harmonic.tex
pdflatex -interaction=nonstopmode -halt-on-error \
  direct_inverse_harmonic.tex
```

The first program needs Python, SymPy, mpmath, and matplotlib; the symbolic program needs SymPy. The included Makefile provides equivalent targets. The PDF can be rebuilt without rerunning the calculations because the figure and generated table are included. There are no repository-relative style dependencies. The LaTeX preamble lists the required packages, including `newtxtext` and `newtxmath`.

B Provenance and statement-level status

B.1 The inspected repository checkpoint

The pinned snapshot is

a795fcffa4b3ece22761d81fbed3c43be8b81795.

The main source for the unresolved question is

Analysis/Transseries/docs/series-and-transseries/Inverse_Harmonic_Stokes_Transport/inverse_harmonic_transseries.tex.

The inspected portions include the normalization and coefficient formulas, the forward-truncation error theorem and proof, and the further questions and references: source ranges 1–280, 430–700, and 948–1119. The Analysis/Transseries/README.md and the series-and-transseries group README were also consulted. The exact source label answered is `prob:direct`; the existing comparison theorem is `thm:optimal`. These anchors are more stable than PDF page numbers.

At this checkpoint, the predecessor is one of the unmerged research packages listed beside the canonical and combinatorial volumes. Its claims are not represented by the repository as newly Lean-formalized. This article does not infer verification from inclusion in the repository.

B.2 Status ledger

Statement or component	Status and dependency
Formal coefficient formula	Inherited from the companion; rederived in proposition 2.1 for normalization.
Sectorial inverse	Proved here using a near-identity Rouché argument and classical sectorial digamma asymptotics. No global injectivity premise.
Positive reflected density	Proved in theorem 3.2 from the exact reflection identity and actual inverse branches.
Exterior dispersion and exact remainder	Proved in theorems 4.1 and 4.3 ; the finite-circle correction is retained.
Eventual enveloping	Proved in theorem 5.1 , uniformly on an existential rectangle; explicit numerical thresholds not supplied.
Sharp direct error	Proved to all algebraic orders in theorem 6.1 ; B_1^- and B_2^- displayed and symbolically checked.
Late coefficients	Independent analytic recovery of the predecessor's large-order result, not a separate novelty claim.
Wider windows and comparison	Proved in theorems 7.2 and 7.4 and corollary 8.2 .
Six sign certificates	Exact rational arithmetic checks using the proven forward integral bound; no floating-point digamma premise.
Numerical asymptotic tests	Finite 190-digit diagnostics at the recorded inputs, not proofs of uniform statements.
Lean and global priority	Neither a new Lean verification nor established global publication priority is asserted.

References

- [1] *ProveIt* research collection, *Exponential Accuracy and Stokes Transport for Inverse Harmonic Transseries*, research note prepared for Vladimir Reshetnikov, 29 September 2026. Pinned snapshot and inspected source ranges in section B; especially the problem `prob:direct` and theorem `thm:optimal`. https://github.com/VladimirReshetnikov/ProveIt/blob/a795fcffa4b3ece22761d81fbed3c43be8b81795/Analysis/Transseries/docs/series-and-transseries/Inverse_Harmonic_Stokes_Transport/inverse_harmonic_transseries.tex.
- [2] *ProveIt* research collection, *Combinatorial Transseries and Their Inverses: Gamma quotients, finite exponential sums, moment sectors, moving saddles, arithmetic sheets, and q-products*, same pinned snapshot; the harmonic-inverse coefficient formula labelled `t2:thm:harmonic-inverse`, also reproduced and audited in [1]. https://github.com/VladimirReshetnikov/ProveIt/tree/a795fcffa4b3ece22761d81fbed3c43be8b81795/Analysis/Transseries/docs/series-and-transseries/Combinatorial_Transseries_Inverses.
- [3] Vladimir Reshetnikov, *ProveIt*, transseries project README and series-and-transseries group README, same pinned snapshot, inspected 29 September 2026. <https://github.com/VladimirReshetnikov/ProveIt/blob/a795fcffa4b3ece22761d81fbed3c43be8b81795/Analysis/Transseries/docs/series-and-transseries/README.md>.
- [4] NIST Digital Library of Mathematical Functions, §5.5, *Functional Relations*, especially equations 5.5.4 and 5.5.8, accessed 29 September 2026. <https://dlmf.nist.gov/5.5>.
- [5] NIST Digital Library of Mathematical Functions, §5.9, *Integral Representations*, and §5.15, *Polygamma Functions*, accessed 29 September 2026. <https://dlmf.nist.gov/5.9>; <https://dlmf.nist.gov/5.15>.
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- [7] B. R. Fabijonas and F. W. J. Olver, “On the reversion of an asymptotic expansion and the zeros of the Airy functions,” *SIAM Review* **41**(4) (1999), 762–773. <https://doi.org/10.1137/S0036144598349538>.
- [8] M. Borinsky, “Generating asymptotics for factorially divergent sequences,” *Electronic Journal of Combinatorics* **25**(4) (2018), P4.1. <https://doi.org/10.37236/5999>. Preprint: <https://arxiv.org/abs/1603.01236>.
- [9] G. Nemes, “Proofs of two conjectures on the real zeros of the cylinder and Airy functions,” *SIAM Journal on Mathematical Analysis* **53**(4) (2021), 4328–4349. <https://doi.org/10.1137/21M1396794>. Preprint: <https://arxiv.org/abs/2010.13069>.